AI-Powered Job Portal Manual Setup Guide

Complete guide to run the project manually using Node.js services and Docker Kafka. Follow the steps in order.

1. Install Required Software

Install Node.js LTS:
https://nodejs.org/

Check:
node -v
npm -v

Install Docker Desktop:
https://www.docker.com/products/docker-desktop/

Check:
docker --version

2. Extract Project

Extract job-portal.zip

Structure:

job-portal
■■■ frontend
■■■ services
■■■ auth
■■■ user
■■■ job
■■■ utils
■■■ payment

3. Environment Setup

Add the provided .env files:

services/auth/.env
services/user/.env
services/job/.env
services/utils/.env
services/payment/.env

4. Install Dependencies

Run npm install inside every service folder:

```
cd services/auth
npm install

cd ../user
npm install

cd ../job
npm install

cd ../utils
npm install

cd ../payment
npm install

cd ../../frontend
npm install
```

5. Kafka Setup

Kafka is required for notification and event processing.

```
docker pull apache/kafka:4.1.0

docker run -d --name kafka `
-p 9092:9092 `
-e KAFKA_NODE_ID=1 `
-e KAFKA_PROCESS_ROLES=broker,controller `
-e KAFKA_LISTENER_SECURITY_PROTOCOL_MAP=PLAINTEXT:PLAINTEXT,CONTROLLER:PLAINTEXT `
-e KAFKA_LISTENERS=PLAINTEXT://:9092,CONTROLLER://:9093 `
-e KAFKA_ADVERTISED_LISTENERS=PLAINTEXT://localhost:9092 `
-e KAFKA_CONTROLLER_LISTENER_NAMES=CONTROLLER `
-e KAFKA_CONTROLLER_QUORUM_VOTERS=1@localhost:9093 `
-e KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR=1 `
-e KAFKA_GROUP_INITIAL_REBALANCE_DELAY_MS=0 `
apache/kafka:4.1.0

Check:
docker ps
```

6. Build Services

```
cd services/auth
npm run build

cd ../user
npm run build

cd ../job
npm run build

cd ../utils
npm run build

cd ../payment
npm run build
```

7. Run Services

```
Auth:
cd services/auth
npm run dev

User:
cd services/user
npm run dev

Job:
cd services/job
npm run dev

Utils:
cd services/utils
npm run dev

Payment:
cd services/payment
npm run dev
```

8. Run Frontend

```
cd frontend
npm run dev

Open:
http://localhost:3000
```

9. Startup Order

1. Docker Desktop
2. Kafka Container
3. Auth Service
4. User Service
5. Job Service
6. Utils Service
7. Payment Service
8. Frontend

Ports

```
Frontend : 3000
Auth     : 5001
Job      : 5002
User     : 5003
Utils    : 5004
Payment  : 3030
Kafka    : 9092
```

Next Time Start

```
docker start kafka
Then run services:
```

```
npm run dev
```